

ASSESSMENT TOOL - 4

Short answer Test

1. Define cloud computing and explain any four characteristics that makes it different from traditional on-premise computing.

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the Internet on a pay-as-you-go basis.

Characteristics:

- i) On-demand self-service: Users can access resources whenever needed.
- ii) Broad network access: Services can be accessed from any device with Internet.
- iii) Resource pooling: Multiple users share the same infrastructure securely.
- iv) Rapid elasticity: Resources can be increased or decreased based on demand.

Example:

Amazon Web Services (AWS) allows users to create virtual servers within minutes.

2. Trace the evolution of cloud computing from hardware evolution to Internet Software evolution, and show how these changes enabled modern cloud platforms.

cloud computing evolved from physical servers, where each application required dedicated hardware. Later, virtualization enabled multiple virtual machines on a single server, improving resource utilization. The growth of the Internet and web applications allowed software to be delivered online instead of being installed locally. This led to cloud platforms like AWS, Microsoft Azure, and Google cloud, which provide scalable, reliable, and cost-effective computing services worldwide.

- 3) Explain the role of server virtualization in cloud computing. Difference between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization allows multiple virtual machines (VMs) to run on a single physical server using a hypervisor. It improves hardware utilization, reduces costs, and enables quick deployment of cloud resources.

Difference:

- * Virtualization is a technology that creates virtual machines.
- * Cloud Computing is a service model that delivers computing resources over the Internet.

Example:

VMware provides virtualization, while AWS provides cloud services.

4.) Discuss the role of data centers in delivering cloud services.

A data center is a facility that contains servers, storage, networking devices, power, and cooling systems to support cloud services. It stores applications and data, provides high availability, ensures security through encryption and firewalls, and supports scalability by adding resources when needed. Modern cloud providers like Google Cloud and Microsoft Azure operate global data centers to deliver reliable cloud services.

5) Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

MODEL	DESCRIPTION	EXAMPLE.
IaaS	Provides virtual servers, storage, and networking	AWS EC2
PaaS	Provides a platform to develop and deploy application	Google App Engine
SaaS	Provides ready-to-use Software Over the Internet	Google Workspace (Gmail)
XaaS	Refers to any cloud-based services	Microsoft Azure AI Services

Difference:

IaaS provides infrastructure, PaaS provides a development platform, SaaS provides complete software, and XaaS includes all cloud-based services.